

# MS&E 228 (Winter 2026) — Lecture 18 Notes

## Dynamic Treatment Effects

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### 1 Introduction: From Static to Sequential Interventions

So far in our study of causal inference, we have primarily focused on **static** or **single-shot interventions**. In this paradigm, a unit is exposed to a treatment at a single point in time, and we are interested in its effect on a subsequent outcome. While this framework is powerful and widely applicable, many real-world scenarios involve a more complex, sequential process where units are treated multiple times over an extended period.

This chapter ventures into this dynamic world, introducing the framework of **Dynamic Treatment Regimes (DTRs)**, also known as adaptive treatment strategies. In a DTR, the treatment administered at any given stage can depend on the entire history of the unit, including its baseline characteristics, prior treatments, and intermediate outcomes. This sequential, adaptive nature of decision-making is ubiquitous in fields like healthcare, marketing, operations, and economics. The goal of this lecture is to introduce this topic, explain why it requires its own dedicated methodology, and generalize the doubly robust and DML frameworks to this new setting.

#### 1.1 Motivating Examples of Sequential Decision-Making

The need to understand and evaluate sequences of interventions arises in numerous domains. Consider the following scenarios:

- **Healthcare:** A cancer patient's treatment is rarely a single event. They might first receive chemotherapy, then surgery, then immunotherapy, and then radiation therapy. These treatments are typically administered in a **reactive or adaptive manner**. For example, a physician administers chemotherapy and observes the patient's response. If the response is positive, they may proceed to surgery; if not, they might turn to immunotherapy. The treatment in the next period depends on both past treatments and the patient's current state. A key causal question is not just about a single drug, but about an entire treatment *pathway*. For instance, what would be the average survival rate if all patients were first treated with

immunotherapy and then with chemotherapy? Such problems arise frequently in healthcare. For instance, a recent study by Szmulewicz et al. (2023) [1] used the methods we will discuss to evaluate different dynamic treatment strategies for Major Depressive Disorder using data from the STAR\*D program, finding that a guidelines-based adaptive strategy yielded the highest remission rates compared to simpler monotherapy or dual-therapy approaches. The ability to evaluate such dynamic policies allows us to discover improved treatment pathways that would otherwise remain hidden.

- **Marketing:** A customer’s journey on an e-commerce platform is a sequence of touchpoints. A firm might first show a customer a banner ad, observe their browsing behavior (e.g., which products they view, whether they add items to their cart), and then decide to offer a targeted discount. This is a sequence of interventions, administered in an adaptive manner. A common goal in marketing is to understand **multi-touch attribution**, which assesses the causal effect of each intervention. For example, what would have been the outcome if we had only shown the ad but not offered the discount?
- **B2B Operations:** In the world of B2B enterprise software, a company like Microsoft might offer a sequence of investments to encourage a customer to adopt its cloud platform. In the first year, they might provide Azure credits to facilitate onboarding. Then, depending on the customer’s cloud adoption over that year, they might assign a dedicated cloud specialist in the second year. The question for the business is to understand the causal effect of this sequence of investments on long-term customer spend, asking counterfactuals like, “What would have happened if we had not assigned a sales specialist?” [2].
- **Labor Economics:** Government-sponsored job training programs often span multiple periods. An unemployed individual might be offered a resume-building workshop in the first period. At the end of this first year, they must decide whether to enroll in the second year. The decision to continue might depend on intermediate outcomes, such as whether they found a job in the interim. The policy question is to evaluate the effect of different durations of the intervention, for instance, the effect of just one year of treatment versus two years of treatment [3].

In all these cases, the treatment in a later period depends on the history of treatments and intermediate outcomes. This creates a significant challenge for causal inference that cannot be solved by the methods we have studied so far.

### Discussion: The Scope of Dynamic Problems

The DTR framework is broadly applicable. What other domains could you imagine such dynamic decision making problems arise? Can you think about domains such as education or sustainability? Have you encountered yourself any problem that you could fit into this paradigm?

## 2 A Mathematical Abstraction: The Two-Period Model

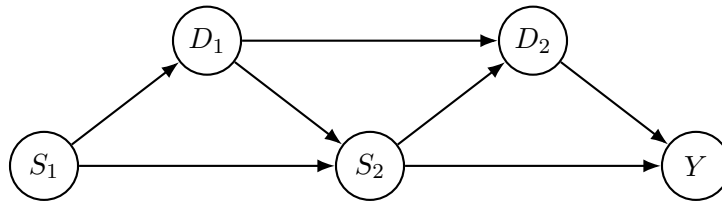
To formally reason about sequential interventions, we can abstract the scenarios described above into a general mathematical framework. For clarity, we will focus on a **two-period model**, as this is sufficient to expose the core identification challenges that distinguish dynamic settings from static ones. The principles we develop here will generalize to settings with more than two time periods.

### 2.1 Notation and The Causal DAG

We model the process for a single unit over two periods with the following set of variables:

- $S_1$ : The initial, pre-treatment state of the unit. This vector includes all baseline covariates, such as demographics or initial health status.
- $D_1$ : The treatment administered in the first period. In an observational study, the choice of  $D_1$  may depend on the initial state  $S_1$ .
- $S_2$ : The state of the unit in the second period. This state is affected by both  $S_1$  and  $D_1$ . It is crucial to note that  $S_1$  and  $S_2$  can be qualitatively very different; for instance,  $S_1$  could be demographics, while  $S_2$  could be a measure of disease progression or economic status after the first intervention. In principle,  $S_2$  can be thought of as a summary of the unit's entire history, potentially including  $S_1$  and  $D_1$  as a subvector.
- $D_2$ : The treatment administered in the second period. The choice of  $D_2$  can depend on the entire history up to that point, which is summarized by the state  $S_2$ .
- $Y$ : The final outcome of interest, which is affected by the second-period state  $S_2$  and treatment  $D_2$ .

This sequence of events gives rise to the causal Directed Acyclic Graph (DAG) shown below. The graph illustrates the flow of causal influence over time.



An important feature of this model is the dual role of the intermediate state  $S_2$ . It is both a **mediator** of the effect of the first treatment (it lies on the causal path  $D_1 \rightarrow S_2 \rightarrow Y$ ) and a **confounder** for the effect of the second treatment (it is a common cause of  $D_2$  and  $Y$ ). This dual role is the central challenge in identifying causal effects in dynamic settings.

## 2.2 The Estimand: Mean Outcome Under a Fixed Joint Intervention

Our primary goal is to estimate the mean potential outcome under a **fixed joint intervention** on the entire sequence of treatments. For our two-period model, the estimand of interest is:

$$\theta = \mathbb{E}[Y(d_1, d_2)]$$

This quantity represents the expected outcome if, hypothetically, we were to intervene and set the first treatment to a specific value  $d_1$  for all units, and subsequently set the second treatment to a specific value  $d_2$  for all units, regardless of their intermediate state  $S_2$ . Let’s map our motivating examples to this abstraction:

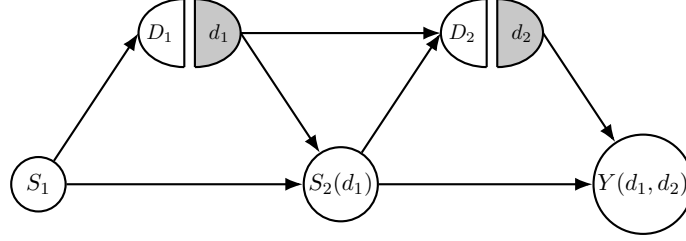
- **Healthcare (STAR\*D):**  $S_1$  is patient baseline characteristics,  $D_1$  is the first-line therapy for depression (e.g., citalopram),  $S_2$  is patient status after some months (remission status, side effects),  $D_2$  is the second-line therapy, and  $Y$  is depression remission at one year. We want to estimate quantities such as  $\mathbb{E}[Y(\text{citalopram}, \text{bupropion})]$ .
- **Marketing (LTV):**  $S_1$  is customer demographics,  $D_1$  is the first marketing action (e.g., banner ad),  $S_2$  is the customer’s state after the action (pages visited, items in cart),  $D_2$  is the second action (e.g., 10% discount), and  $Y$  is the customer’s 2-year revenue. We want to estimate  $\mathbb{E}[Y(\text{banner}, \text{discount})]$ .

### Fixed Interventions vs. Adaptive Data

It is important to distinguish the *nature of the intervention* we wish to evaluate from the *nature of the observational data*. The observational data may well have been generated by an **adaptive** process, where the treatment  $D_2$  administered by a physician or a system depended on the observed state  $S_2$ . For instance, the arrow  $D_1 \rightarrow D_2$  in the DAG indicates that the first treatment might have a direct influence on the choice of the second, which is a hallmark of an adaptive process in the data-generating mechanism.

However, our causal question is about the effect of a **fixed** policy. We are using the potentially adaptive observational data to answer a “what if” question about a non-adaptive, fixed intervention (e.g., “what if we give everyone drug A, then drug B?”). We will later see how to evaluate counterfactual adaptive policies as well (e.g., “what if we give drug A, and then give drug B only if the patient responds?”), but for now, the target of estimation is a fixed sequence.

To reason about this joint intervention, we can draw the Single World Intervention Graph (SWIG), which makes the counterfactual variables explicit. The SWIG below corresponds to the intervention  $\text{do}(D_1 = d_1, D_2 = d_2)$ .



In this graph,  $S_2(d_1)$  is the potential intermediate state that would be realized if the first treatment were set to  $d_1$ , and  $Y(d_1, d_2)$  is the final potential outcome under the full joint intervention. Our goal is to estimate the expectation of this final outcome.

### 3 The Identification Challenge: Time-Varying Confounding

Having formalized our estimand, we now confront the central problem of identification. Can we estimate  $\mathbb{E}[Y(d_1, d_2)]$  from observational data using the methods we have developed so far, such as the g-formula based on conditioning? The answer, it turns out, is no. The temporal structure of the problem introduces a unique challenge known as **time-varying confounding**.

#### 3.1 Why Standard Conditioning Fails

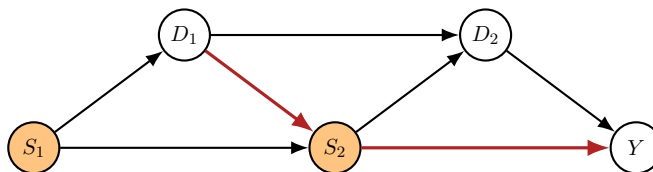
Let's explore two intuitive but incorrect approaches to identification.

**Attempt 1: Condition on the Full History of States.** A naive strategy might be to treat the pair of treatments  $(D_1, D_2)$  as a single, two-dimensional static treatment and to condition on the full history of observed states,  $(S_1, S_2)$ . This would suggest the following identification formula:

$$\mathbb{E}[Y|D_1 = d_1, D_2 = d_2, S_1, S_2] \stackrel{?}{=} \mathbb{E}[Y(d_1, d_2)|S_1, S_2]$$

This approach is fundamentally flawed. The intermediate state  $S_2$  is a **post-treatment variable** with respect to the first treatment  $D_1$ . As we have learned, conditioning on a post-treatment variable can induce bias. In this specific DAG,  $S_2$  is a mediator of the effect of  $D_1$  on  $Y$ . By conditioning on  $S_2$ , we block the very causal pathway we are trying to measure.

To see this clearly, consider a simple thought experiment where the second treatment has no effect on the outcome, but  $D_1$  affects  $Y$  only through  $S_2$ . If we condition on both  $S_1$  and  $S_2$ , we will estimate a zero effect for  $D_1$ , because once we fix  $S_2$ , the path from  $D_1$  to  $Y$  is blocked. We would wrongly conclude that the first treatment is ineffective.

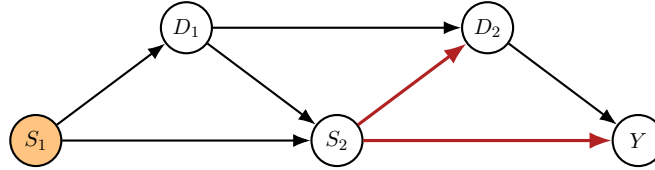


Furthermore, from the perspective of the SWIG, the conditional exogeneity assumption that  $Y(d_1, d_2) \perp\!\!\!\perp (D_1, D_2) \mid S_1, S_2$  is not even checkable, because the variable  $S_2$  does not exist on the SWIG; only its counterfactual version  $S_2(d_1)$  does. This underscores a key principle: if an identification assumption cannot be verified on the SWIG, it should not be used.

**Attempt 2: Condition Only on the Pre-Treatment State.** If conditioning on post-treatment variables is problematic, what if we only condition on the baseline, pre-treatment state  $S_1$ ? This would suggest:

$$\mathbb{E}[Y \mid D_1 = d_1, D_2 = d_2, S_1] \stackrel{?}{=} \mathbb{E}[Y(d_1, d_2) \mid S_1]$$

This approach also fails. While conditioning on  $S_1$  is sufficient to identify the effect of the first treatment  $D_1$  (i.e.,  $Y(d_1) \perp\!\!\!\perp D_1 \mid S_1$  holds), it is *not* sufficient for the second treatment  $D_2$ . As shown in the DAG below, the intermediate state  $S_2$  is a common cause of both  $D_2$  and  $Y$ . By conditioning only on  $S_1$ , we leave the confounding path  $D_2 \leftarrow S_2 \rightarrow Y$  unblocked.



This unblocked path means that the assumption of conditional exogeneity for the joint treatment,  $Y(d_1, d_2) \perp\!\!\!\perp (D_1, D_2) \mid S_1$ , is violated. We are caught in a no-win situation: to identify the effect of  $D_2$ , we need to condition on  $S_2$ , but conditioning on  $S_2$  biases the estimated effect of  $D_1$ .

### The Core Challenge: A Mediator-Confounder

The fundamental difficulty in dynamic settings is the dual role played by intermediate, time-varying variables. The state  $S_2$  is simultaneously:

- A **mediator** of the effect of  $D_1$  on  $Y$ .
- A **confounder** of the effect of  $D_2$  on  $Y$ .

This type of variable is often called a **time-varying confounder affected by past treatment**. Because of this dual role, simple conditioning on any static set of variables is insufficient for identification. We need a more sophisticated, dynamic approach.

## 3.2 A New Strategy: What Can We Identify by Conditioning?

When we reach an identification impasse, a productive strategy is to step back and ask: what causal effects *can* we identify with the tools we have? While we cannot identify the joint effect of  $(D_1, D_2)$  by simple conditioning, we *can* identify the effects of unilateral interventions.

- **Effect of  $D_2$  on  $Y$ :** We can identify the causal effect of the second-period treatment on the final outcome by conditioning on the intermediate state  $S_2$ . Given  $S_2$ , there are no unblocked backdoor paths between  $D_2$  and  $Y$ .
- **Effect of  $D_1$  on  $S_2$ :** We can identify the causal effect of the first-period treatment on the intermediate state by conditioning on the baseline state  $S_1$ . Given  $S_1$ , there are no unblocked backdoor paths between  $D_1$  and  $S_2$ .

This observation is the key that unlocks the identification puzzle. The overall strategy, which we will formalize in the next section, is to chain these identifiable effects together. Roughly speaking, the approach is as follows: we will first estimate the effect of  $D_2$  on  $Y$  as a function of the state  $S_2$ , i.e., the quantity  $\mathbb{E}[Y(d_2) \mid S_2]$ . This gives us a model that can predict the counterfactual outcome  $Y(d_2)$  for any unit, given its state  $S_2$ , if we were to intervene only in the second period. Then, we will use the fact that (in a world where we have already intervened in the second period treatment) the effect of  $D_1$  on the final outcome  $Y(d_2)$  is fully mediated by  $S_2$ . Therefore, to find the effect of  $D_1$  on  $Y(d_2)$  (which would be the variable  $Y(d_1, d_2)$ ), it suffices to find the effect of  $D_1$  on our *forecast* of  $Y(d_2)$  from  $S_2$ . Since we can identify the effect of  $D_1$  on  $S_2$ , by conditioning on  $S_1$ , we can also identify its effect on this forecast, by conditioning on  $S_1$ . By combining these two steps (though not linearly), we can recover the total effect of the joint intervention.

## 4 Identification via Backwards Recursion: Sequential G-Formula

Since simple conditioning fails, we need a strategy that can dynamically account for the evolving structure of the problem. The solution, first proposed by Robins (1986) [4], is a powerful and elegant identification strategy known as the **sequential g-formula** or **g-computation formula**. The core idea is to solve the identification problem via **backwards recursion**.

Instead of trying to identify the joint effect of  $(D_1, D_2)$  at once, we break the problem down, starting from the end and working our way backward in time.

### 4.1 Step 1 (The Last Period): Identifying the Effect of $D_2$

Let's first consider a simpler problem: what is the effect of an intervention only on the *last* treatment,  $D_2$ ? If we only intervene on  $D_2$ , the causal DAG shows that the intermediate state  $S_2$  is a sufficient set of covariates to control for confounding between  $D_2$  and  $Y$ . The path  $D_2 \leftarrow S_2 \rightarrow Y$  is blocked by conditioning on  $S_2$ .

Therefore, under the assumption of **sequential exchangeability**, we can identify the conditional mean of the potential outcome  $Y(d_2)$ :

$$Y(d_2) \perp\!\!\!\perp D_2 \mid S_2$$

This allows us to identify the expected outcome under a second-period intervention, conditional on the state  $S_2$ , using a standard backdoor adjustment. We define a function, which we will call  $Q_2$ ,

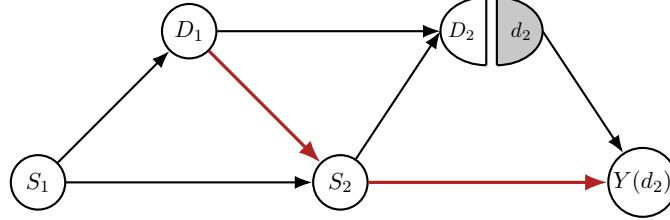
that represents this conditional expectation:

$$Q_2(S_2, d_2) := \mathbb{E}[Y(d_2)|S_2] = \mathbb{E}[Y|S_2, D_2 = d_2]$$

This function  $Q_2(s_2, d_2)$  can be interpreted as a **forecast** of the final outcome  $Y$  for a unit with intermediate state  $s_2$ , if we were to intervene and force that unit to receive treatment  $d_2$  in the second period. Since the statistical quantity  $\mathbb{E}[Y | S_2, D_2]$  is a standard conditional expectation, we can estimate it from data using any regression model (e.g., machine learning) that predicts  $Y$  from the features  $(S_2, D_2)$ . Then for any sample with state  $S_2$  we can simply apply this predictive model with inputs  $(S_2, d_2)$  (i.e., throw away its natural treatment value  $D_2$  and replace it with the constant  $d_2$ ), to get the predicted mean counterfactual outcome  $\mathbb{E}[Y(d_2) | S_2]$ .

## 4.2 Step 2 (The First Period): Recursion and Full Mediation

Now for the crucial insight. Consider the world where we have already intervened and fixed the second-period treatment to  $d_2$ . In this hypothetical world, the outcome is  $Y(d_2)$ . If we were to intervene and fix  $D_1$  to  $d_1$ , then we would get the interventional outcome  $Y(d_1, d_2)$ , which is what we are after. So we are essentially asking for the effect of intervening on  $D_1$  on the outcome  $Y(d_2)$ . What is the effect of the first-period treatment,  $D_1$ , on this outcome? Looking at the SWIG, we can see that the effect of  $D_1$  on  $Y(d_2)$  is **fully mediated** by the intermediate potential state  $S_2$ .

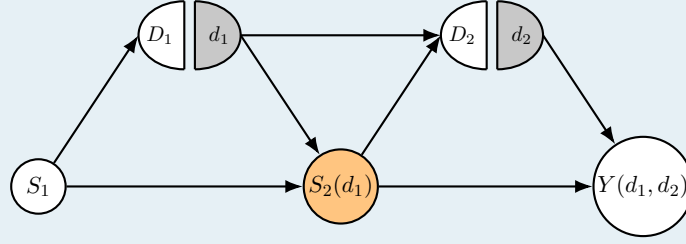


Because of this full mediation, the effect of intervening on  $D_1$  on the final outcome  $Y(d_2)$  is equivalent to the effect of intervening on  $D_1$  on the *forecast*  $Y_{adj}$  of that outcome; forecasted from  $S_2$ . The forecast, as we already defined, is the function  $Y_{adj} := Q_2(S_2, d_2)$ . Thus we can write:

$$\mathbb{E}[Y(d_1, d_2) | S_1] = \mathbb{E}[Y_{adj}(d_1) | S_1] = \mathbb{E}[Q_2(S_2(d_1), d_2) | S_1]$$



### Formal Argument for Fully Mediated Effect



$$\begin{aligned}
 \mathbb{E}[Y(d_1, d_2) \mid S_1] &= \mathbb{E}[\mathbb{E}[Y(d_1, d_2) \mid S_2(d_1), S_1] \mid S_1] && \text{(tower rule)} \\
 &= \mathbb{E}[\mathbb{E}[Y(d_1, d_2) \mid S_2(d_1), D_1 = d_1] \mid S_1] && (d\text{-separation in SWIG}) \\
 &= \mathbb{E}[\mathbb{E}[Y(D_1, d_2) \mid S_2(d_1), D_1 = d_1] \mid S_1] && \text{(consistency)} \\
 &= \mathbb{E}[\mathbb{E}[Y(d_2) \mid S_2 = S_2(d_1), D_1 = d_1] \mid S_1] && \text{(consistency)} \\
 &= \mathbb{E}[\mathbb{E}[Y(d_2) \mid S_2 = S_2(d_1)] \mid S_1] && (d\text{-separation in SWIG fix}(D_2 = d_2))
 \end{aligned}$$

This allows us to define a new, **adjusted outcome variable**  $Y_{adj}$ . This variable is the output of our forecasting model from Step 1, evaluated for each unit using their observed intermediate state  $S_2$  but with the second-period treatment fixed to our target value  $d_2$ . The problem of estimating  $\mathbb{E}[Y(d_1, d_2)]$  has now been reduced to estimating the effect of  $D_1$  on this new adjusted outcome, i.e.,  $\mathbb{E}[Y_{adj}(d_1)]$ .

Crucially, we can identify *this* effect by conditioning, since we have the conditional ignorability property:

$$Y_{adj}(d_1) \perp\!\!\!\perp D_1 \mid S_1$$

This holds since  $Y_{adj}$  is just a function of  $S_2$ , and the effect of  $D_1$  on  $S_2$  is unconfounded given  $S_1$

$$S_2(d_1) \perp\!\!\!\perp D_1 \mid S_1$$

Hence, any function of  $S_2$  also satisfies the same independence property, i.e.,

$$Q_2(S_2(d_1), d_2) \perp\!\!\!\perp D_1 \mid S_1$$

Then we note that  $Y_{adj}(d_1) = Q_2(S_2(d_1), d_2)$ . Therefore, we can again use identification by conditioning. We define a second function,  $Q_1$ , which models the conditional expectation of our adjusted outcome:

$$Q_1(S_1, d_1) := \mathbb{E}[Y_{adj}(d_1) \mid S_1] = \mathbb{E}[Y_{adj} \mid S_1, D_1 = d_1] = \mathbb{E}[Q_2(S_2, d_2) \mid S_1, D_1 = d_1]$$

This function  $Q_1(s_1, d_1)$  represents the expected final outcome for a unit with initial state  $s_1$ , if that unit were to receive treatment  $d_1$  in the first period and we subsequently follow the policy

of applying treatment  $d_2$  in the second period, i.e.,  $Q_1(s_1, d_1) = \mathbb{E}[Y(d_1, d_2) \mid S_1 = s_1]$ . Thus to estimate the final mean counterfactual we care about, we simply average these final values:

$$\mathbb{E}[Y(d_1, d_2)] = \mathbb{E}[\mathbb{E}[Y(d_1, d_2) \mid S_1]] = \mathbb{E}[Q_1(S_1, d_1)]$$

### A Clinical Walkthrough of Backwards Recursion

Let's walk through this with a clinical example. Suppose we have observational data on cancer patients over two years. Our goal is to evaluate a policy of giving all patients immunotherapy in year 1, followed by chemotherapy in year 2.

1. **Step 1 (Year 2):** We first go to the second year. We build a model to predict long-term survival ( $Y$ ) based on the patient's state at the end of the first year ( $S_2$ , e.g., tumor size, disease stage) and the second-year treatment ( $D_2$ , e.g., chemotherapy or immunotherapy). This is our  $\hat{Q}_2$  model. Because we assume conditional exogeneity given  $S_2$ , this model has a causal interpretation. For any patient, it can tell us their expected survival if they get chemo, and their expected survival if they get immunotherapy, given their state.
2. **Step 2 (Adjusting the Outcome):** Our policy is to always give chemotherapy in year 2. So, for every patient in our dataset, we use our  $\hat{Q}_2$  model to predict their survival *as if* they had received chemotherapy, regardless of what they actually got. This gives us an adjusted outcome,  $Y_{adj,2} = \hat{Q}_2(S_2, \text{chemotherapy})$ , for every patient.
3. **Step 3 (Year 1):** Now, we want to know the effect of giving immunotherapy in year 1 (on top of giving chemo in year 2). We need to estimate how first-year immunotherapy changes the patient's state  $S_2$ , which in turn changes the forecasted outcome from our second-stage model. Equivalently, we can directly estimate the effect of the first-year treatment on the adjusted outcome,  $Y_{adj,2}$ . We build a new model,  $\hat{Q}_1$ , that predicts  $Y_{adj,2}$  using the initial state  $S_1$  and the first-year treatment  $D_1$ . Because we assume conditional exogeneity of  $D_1$  on  $S_2$  given  $S_1$ , this model has a causal interpretation. For any patient, it can tell us their expected survival if they get chemo, and their expected survival if they get immunotherapy in the first year, given their initial state (and continuing with chemo in the second year).
4. **Step 4 (Final Estimate):** Our policy is to give immunotherapy in year 1. We use our  $\hat{Q}_1$  model to predict the outcome for every patient *as if* they had received immunotherapy in year 1. This gives us a final adjusted outcome,  $Y_{adj,1} = \hat{Q}_1(S_1, \text{immunotherapy})$ . To get the average counterfactual outcome for the entire population, under the sequence of treatments (immunotherapy, chemo), we simply average these final adjusted outcomes across all individuals, i.e.  $\mathbb{E}[Y_{adj,1}]$ .

### 4.3 Putting It All Together: The G-Formula

#### The Sequential G-Formula

Under the assumption of sequential conditional ignorability:

$$Y(D_1, d_2) \perp\!\!\!\perp D_2 \mid S_2 \quad S_2(d_1) \perp\!\!\!\perp D_1 \mid S_1$$

and complete mediation of effect of  $D_1$  via  $S_2$ :

$$Y(d_2) \perp\!\!\!\perp (S_1, D_1) \mid S_2$$

and assuming sequential overlap  $\Pr(D_t = d_t \mid S_t) > 0$  a.s., the mean potential outcome under a fixed joint intervention  $(d_1, d_2)$  is identified by the nested conditional expectation:

$$\mathbb{E}[Y(d_1, d_2)] = \mathbb{E}_{S_1} \left[ \mathbb{E}_{S_2 \mid S_1, D_1 = d_1} [\mathbb{E}[Y \mid S_2, D_2 = d_2]] \right]$$

This can be written recursively using the Q-functions as:

$$\mathbb{E}[Y(d_1, d_2)] = \mathbb{E}_{S_1} [Q_1(S_1, d_1)]$$

$$Q_1(s_1, d_1) = \mathbb{E}[Q_2(S_2, d_2) \mid S_1 = s_1, D_1 = d_1]$$

$$Q_2(s_2, d_2) = \mathbb{E}[Y \mid S_2 = s_2, D_2 = d_2]$$

#### Connection to Reinforcement Learning: Fitted Q-Iteration

This backwards recursive procedure is known in the reinforcement learning literature as **Fitted Q-Iteration**. The Q-functions,  $Q_t(s_t, d_t)$ , represent the expected cumulative future reward (in our case, just the final outcome  $Y$ ) starting from state  $s_t$ , taking action  $d_t$ , and following a fixed policy thereafter. The algorithm works by:

1. Estimating the Q-function for the final period,  $Q_T$ .
2. Using the predictions of the fitted model  $\hat{Q}_T$  as the labels (or targets) for estimating the Q-function of the previous period,  $Q_{T-1}$ .
3. Repeating this process, iterating backwards until the first period's Q-function,  $Q_1$ , is estimated.

Our identification strategy is a direct application of this principle to the problem of causal inference.

### Statistical Problem of Many Treatment Sequences

It is important to note that while we fit a single Q-function for every period, there exists an exponential number of possible treatment sequences. Each individual sequence might be observed very infrequently in the data, which poses a statistical challenge unless further modeling assumptions are made.

### Alternative Formal Derivation of the G-Formula

The identification result can also be formally shown through an alternative sequence of steps using the law of total expectation (tower rule), d-separation on the SWIG, and the consistency assumption.

$$\begin{aligned}
\mathbb{E}[Y(d_1, d_2)] &= \mathbb{E}[\mathbb{E}[Y(d_1, d_2) \mid S_1]] && \text{(Tower Rule)} \\
&= \mathbb{E}[\mathbb{E}[Y(d_1, d_2) \mid S_1, D_1 = d_1]] && (Y(d_1, d_2) \perp\!\!\!\perp D_1 \mid S_1) \\
&= \mathbb{E}[\mathbb{E}[Y(D_1, d_2) \mid S_1, D_1 = d_1]] && \text{(Consistency)} \\
&= \mathbb{E}[\mathbb{E}[Y(d_2) \mid S_1, D_1 = d_1]] && \text{(Consistency)} \\
&= \mathbb{E}[\mathbb{E}[\mathbb{E}[Y(d_2) \mid S_2, S_1, D_1 = d_1] \mid S_1, D_1 = d_1]] && \text{(Tower Rule)} \\
&= \mathbb{E}[\mathbb{E}[\mathbb{E}[Y(d_2) \mid S_2] \mid S_1, D_1 = d_1]] && (Y(d_2) \perp\!\!\!\perp (S_1, D_1) \mid S_2) \\
&= \mathbb{E}[\mathbb{E}[\mathbb{E}[Y(d_2) \mid S_2, D_2 = d_2] \mid S_1, D_1 = d_1]] && (Y(d_2) \perp\!\!\!\perp D_2 \mid S_2) \\
&= \mathbb{E}[\mathbb{E}[\mathbb{E}[Y \mid S_2, D_2 = d_2] \mid S_1, D_1 = d_1]] && \text{(Consistency)}
\end{aligned}$$

The key assumptions used here are sequential ignorability

$$Y(d_1, d_2) \perp\!\!\!\perp D_1 \mid S_1 \quad Y(d_2) \perp\!\!\!\perp D_2 \mid S_2$$

and that  $S_2$  is a complete mediator of the effect of  $(S_1, D_1)$  on the potential outcome  $Y(d_2)$ .

$$Y(d_2) \perp\!\!\!\perp (S_1, D_1) \mid S_2$$

## 5 Estimation: From Plug-in to Doubly Robust Methods

The sequential g-formula provides a clear identification strategy, but its nested structure presents challenges for estimation, particularly regarding statistical inference (i.e., calculating standard errors and confidence intervals).

### 5.1 A Plug-in G-Computation Algorithm

The recursive definition of the g-formula immediately suggests a "plug-in" estimation algorithm:

### Algorithm: Plug-in G-Computation for a Two-Period Model

1. **Estimate the final-stage Q-function,  $Q_2$ :** Train a machine learning model (e.g., random forest, gradient boosting) to predict the outcome  $Y$  using the features  $(S_2, D_2)$ . Let the fitted model be  $\hat{Q}_2(s_2, d_2)$ .
2. **Generate pseudo-outcomes:** For each observation  $i$  in the dataset, compute an adjusted outcome  $\hat{Y}_{i,adj,2}$  by evaluating the fitted model  $\hat{Q}_2$  using the observed intermediate state  $S_{2,i}$  but with the second-period treatment fixed to the target value  $d_2$ :

$$\hat{Y}_{i,adj,2} = \hat{Q}_2(S_{2,i}, d_2)$$

3. **Estimate the first-stage Q-function,  $Q_1$ :** Train another machine learning model to predict the pseudo-outcomes  $\hat{Y}_{adj,2}$  using the features  $(S_1, D_1)$ . Let the fitted model be  $\hat{Q}_1(s_1, d_1)$ .
4. **Generate final predictions:** For each observation  $i$ , compute the final predicted potential outcome by evaluating  $\hat{Q}_1$  using the observed baseline state  $S_{1,i}$  but with the first-period treatment fixed to the target value  $d_1$ :

$$\hat{Y}_{i,adj,1} = \hat{Q}_1(S_{1,i}, d_1)$$

5. **Estimate the mean potential outcome:** The final estimate is the sample average of these predictions:

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n \hat{Y}_{i,adj,1}$$

To calculate a treatment effect, such as the average treatment effect of always treating versus never treating, this entire procedure must be run multiple times for different treatment combinations (e.g., once for  $(d_1, d_2) = (1, 1)$  and once for  $(d_1, d_2) = (0, 0)$ ). Other causal contrasts of interest include:

- $Y(1, 1) - Y(1, 0)$ : The marginal effect of the second-period treatment, given treatment in the first period.
- $Y(1, 0) - Y(0, 0)$ : The effect of only treating in the first period.

While this algorithm is intuitive, it suffers from a critical flaw. If we simply calculate the standard deviation of the final predictions and divide by  $\sqrt{n}$ , we will not get valid standard errors. This is for the same reason that the simple  $g$ -formula in the static case did not provide valid inference when using machine learning models. The plug-in estimator  $\hat{\theta}$  is generally not asymptotically normal, and its bias can be large due to the complex, nested way it depends on the ML models. To achieve valid inference, we must once again turn to a **doubly robust** approach.

## 5.2 A Multiply Robust Estimator for Sequential Interventions

To debias the plug-in estimator, we need to add correction terms that account for the estimation error in *each* of the nuisance functions  $Q$ . This requires not only learning the Q-functions but also the **propensity scores** for each period (analogous to the static case):

- $p_1(S_1) = P(D_1 = d_1|S_1)$ : The probability of receiving treatment  $d_1$  in the first period, given baseline state  $S_1$ .
- $p_2(S_2) = P(D_2 = d_2|S_2)$ : The probability of receiving treatment  $d_2$  in the second period, given intermediate state  $S_2$ .

We can build the debiased estimator step-by-step, mirroring our recursive identification strategy.

**Debiasing Step 1: Correcting for  $Q_1$ .** The static g-formula  $\mathbb{E}[g(1, X) - g(0, X)]$  depended on a regression function  $g(D, X) = \mathbb{E}[Y | D, X]$ , and we made it robust by adding a correction term involving the residuals of that regression multiplied by inverse propensity weights (the Horvitz-Thompson weights). We can apply the same principle to the sequential g-formula  $\mathbb{E}[Q_1(S_1, d_1)]$ , which depends on the regression function  $Q_1(S_1, D_1) = \mathbb{E}[Q_2(S_2, d_2) | S_1, D_1]$ . The resulting formula is now insensitive to estimation errors in  $Q_1$ :

Step 1: Debiasing  $Q_1$

$$\underbrace{Q_1(S_1, d_1)}_{\text{Original formula}} + \underbrace{\frac{\mathbb{I}(D_1 = d_1)}{\Pr(D_1 = d_1 | S_1)}}_{\text{Inverse propensity}} \cdot \underbrace{(Q_2(S_2, d_2) - Q_1(S_1, D_1))}_{\text{Residual of regression defining } Q_1}$$

However, this formula is not fully debiased. The correction term still contains  $Q_2$ , which is itself an estimated nuisance function.

**Debiasing Step 2: Correcting for  $Q_2$ .** Since our formula still depends on the nuisance function  $Q_2$ , we can apply the same debiasing logic again. We can add a second correction term that is based on the residuals of the regression defining  $Q_2$ , which is  $Y - Q_2(S_2, D_2)$ . This residual is multiplied by the corresponding Horvitz-Thompson weight for the second period,  $\frac{\mathbb{I}(D_2=d_2)}{\Pr(D_2=d_2|S_2)}$ . However, since the term we are correcting,  $Q_2$ , was already being multiplied by the first-period weights, we must multiply this new correction term by those weights as well. This leads to the final debiased formula.

### Final Dynamic Doubly Robust Formula

$$\begin{aligned}
 Y_{DR}(Q_{1:2}, p_{1:2}) = & Q_1(S_1, d_1) + \frac{\mathbb{I}(D_1 = d_1)}{\Pr(D_1 = d_1 \mid S_1)} (Q_2(S_2, d_2) - Q_1(S_1, D_1)) \\
 & + \underbrace{\frac{\mathbb{I}(D_2 = d_2)}{\Pr(D_2 = d_2 \mid S_2)}}_{\text{New IPS for period 2}} \cdot \underbrace{\frac{\mathbb{I}(D_1 = d_1)}{\Pr(D_1 = d_1 \mid S_1)}}_{\text{IPS already multiplying } Q_2} \cdot \underbrace{(Y - Q_2(S_2, D_2))}_{\text{Residual of regression defining } Q_2}
 \end{aligned}$$

This formula has a property known as **multiple robustness**. The quantity:

$$\theta(Q_{1:2}, p_{1:2}) = \mathbb{E}[Y_{DR}(Q_{1:2}, p_{1:2})]$$

recovers the correct target quantity  $\mathbb{E}[Y(d_1, d_2)]$  if either all  $Q$ -functions  $Q_{1:2}$  are correct or if all propensity functions  $p_{1:2}$  are correct. It does not require all models to be correct simultaneously, making it a very robust procedure. Notably, if all propensity scores are correct and the  $Q$  functions are zero, then the formula simplifies to the product of inverse propensities multiplied by the outcome:

$$\mathbb{E} \left[ \frac{\mathbb{I}(D_2 = d_2)}{\Pr(D_2 = d_2 \mid S_2)} \cdot \frac{\mathbb{I}(D_1 = d_1)}{\Pr(D_1 = d_1 \mid S_1)} \cdot Y \right]$$

which is the analogue of the IPW formula in the dynamic treatment regime. If all  $Q$ -functions are correct, the correction terms vanish in expectation, and we are left with the nested conditional expectation g-formula.

The dynamic doubly robust formula naturally leads to an estimator:

$$\hat{\theta} = \frac{1}{n} \sum_i Y_{DR}(\hat{Q}_{1:2}, \hat{p}_{1:2})$$

To implement this, we use cross-fitting (as in the static case) to estimate all nuisance functions ( $\{\hat{Q}_1, \hat{Q}_2, \hat{p}_1, \hat{p}_2\}$ ) on separate folds of the data to avoid overfitting bias, and then compute the average of the influence function values.

## Dynamic Doubly Robust Estimator: Pseudocode

```
import numpy as np
from sklearn.model_selection import KFold
# Nuisance models: Q2_model, Q1_model (regressors), p1_model, p2_model (classifiers)
psi_values = np.zeros(n)
kf = KFold(n_splits=K)
for train_idx, test_idx in kf.split(S1):
    # Stage 2: fit Q2(S2, D2) -> Y, and propensity p2(S2) -> D2
    Q2_model.fit(np.c_[S2[train_idx], D2[train_idx]], Y[train_idx])
    p2_model.fit(S2[train_idx], D2[train_idx])

    # Stage 1: fit Q1(S1, D1) -> Q2(S2, d2) and propensity p1(S1) -> D1
    Q2_pseudo = Q2_model.predict(np.c_[S2[train_idx], np.full(len(train_idx), d2)])
    Q1_model.fit(np.c_[S1[train_idx], D1[train_idx]], Q2_pseudo)
    p1_model.fit(S1[train_idx], D1[train_idx])

    # Predict on test fold
    Q2_hat = Q2_model.predict(np.c_[S2[test_idx], D2[test_idx]])
    Q1_hat = Q1_model.predict(np.c_[S1[test_idx], D1[test_idx]])
    Q2_hat_d2 = Q2_model.predict(np.c_[S2[test_idx], np.full(len(test_idx), d2)])
    Q1_hat_d1 = Q1_model.predict(np.c_[S1[test_idx], np.full(len(test_idx), d1)])
    p1_hat = p1_model.predict_proba(S1[test_idx].reshape(-1,1))[:, d1]
    p2_hat = p2_model.predict_proba(S2[test_idx])[:, d2]

    # Doubly robust score
    w1 = (D1[test_idx]==d1) / np.clip(p1_hat, 0.01, 1)
    w2 = (D2[test_idx]==d2) / np.clip(p2_hat, 0.01, 1)
    psi_values[test_idx] = Q1_hat_d1 + w1 * (Q2_hat_d2 - Q1_hat)\
        + w2 * w1 * (Y[test_idx] - Q2_hat)

theta_hat = np.mean(psi_values)
se_hat = np.std(psi_values) / np.sqrt(n)
ci = (theta_hat - 1.96 * se_hat, theta_hat + 1.96 * se_hat)
```



## Asymptotic Normality of the Dynamic DR Estimator

Assume sequential conditional ignorability (implied by the DAG)

$$Y(d_2) \perp\!\!\!\perp D_2 \mid S_2 \quad S_2(d_1) \perp\!\!\!\perp D_1 \mid S_1$$

and  $S_2$  is a complete mediator of the effect of  $D_1, S_1$  on  $Y(d_2)$ , i.e.,

$$Y(d_2) \perp\!\!\!\perp (D_1, S_1) \mid S_2.$$

Assume **strict** sequential overlap, i.e.,

$$\Pr(D_t \mid S_t) \in (\epsilon, 1 - \epsilon).$$

Assume the  $Q$  functions and the propensities are consistent  $\text{RMSE}(\hat{Q}_t), \text{RMSE}(\hat{p}_t) \rightarrow 0$  and the following *Product Rate Conditions* hold:

$$\sqrt{n} \left( \text{RMSE}(\hat{Q}_1) \text{RMSE}(\hat{p}_1) + \text{RMSE}(\hat{Q}_2) \text{RMSE}(\hat{p}_1) + \text{RMSE}(\hat{Q}_2) \text{RMSE}(\hat{p}_2) \right) \rightarrow 0,$$

Under mild regularity conditions, the Dynamic DR Estimator with cross-fitting satisfies

$$\sqrt{n}(\hat{\theta}_{\text{DR}} - \mathbb{E}[Y(d_1, d_2)]) \Rightarrow \mathcal{N}(0, \text{Var}(Y_{\text{DR}})).$$

Standard errors can be estimated as:

$$\hat{\text{se}} = \sqrt{\text{Var}_n(\hat{Y}_{\text{DR}})/n}, \quad \text{and} \quad 95\% \text{ CI} = \hat{\theta}_{\text{DR}} \pm 1.96\hat{\text{se}}.$$

## 6 Extensions: Adaptive Policies and Multiple Periods

Our discussion so far has focused on estimating the effect of a *fixed* joint intervention. However, the true power of the dynamic treatment regime framework lies in its ability to evaluate **adaptive policies**, where the treatment at a given stage is a function of the current state.

### 6.1 Evaluating Adaptive Policies

An adaptive policy (or treatment rule) is a set of functions,  $\pi = \{\pi_1, \pi_2, \dots\}$ , where each function  $\pi_t(S_t)$  maps the state at time  $t$  to a specific treatment. For example, the guidelines-based strategy in the STAR\*D trial is an adaptive policy:

- First period: Always start with Citalopram. So,  $\pi_1(S_1) = \text{Citalopram}$ .
- Second period: If the patient has not remitted (a feature of state  $S_2$ ), switch to a new drug; otherwise, augment the current one. This can be written as a deterministic function of  $S_2$ .

For instance,  $\pi_2(S_2) = \text{switch} \cdot \mathbb{I}(\text{no response}) + \text{augment} \cdot \mathbb{I}(\text{partial response})$ .

The goal is to estimate the mean potential outcome if this entire policy were deployed:

$$\mathbb{E}[Y(\pi_1(S_1), \pi_2(S_2(\pi_1(S_1))))].$$

We will be using the short-hand notation:

$$\mathbb{E}[Y(\pi_1, \pi_2)].$$

for the interventional counterfactual under such an adaptive policy.

Fortunately, the entire framework we have developed generalizes to this case with a simple modification. Everywhere a fixed treatment value  $d_t$  appeared in our formulas, we now substitute the policy function  $\pi_t(S_t)$ . For example, the  $Q$  functions should now be recursively defined as:

$$\begin{aligned} Q_2(S_2, D_2) &= \mathbb{E}[Y \mid S_2, D_2] \\ Q_1(S_1, D_1) &= \mathbb{E}[Q_2(S_2, \pi_2(S_2)) \mid S_1, D_1] \end{aligned}$$

and the plug-in g-computation algorithm is modified as follows:

- In Step 2, the pseudo-outcomes are generated using the policy rule:  $\hat{Y}_{i,adj,2} = \hat{Q}_2(S_{2,i}, \pi_2(S_{2,i}))$ .
- In Step 4, the final predictions are generated using the policy rule:  $\hat{Y}_{i,adj,1} = \hat{Q}_1(S_{1,i}, \pi_1(S_{1,i}))$ .

The doubly robust influence function is similarly modified by replacing the fixed  $d_t$  values with the policy functions  $\pi_t(S_t)$  and adjusting the propensity score denominators to reflect the probability of receiving the treatment prescribed by the policy.

#### Doubly Robust Formula for Adaptive Policy $\pi$

$$\begin{aligned} Y_{DR} &= Q_1(S_1, \pi_1(S_1)) \\ &+ \frac{\mathbb{I}(D_1 = \pi_1(S_1))}{\Pr(D_1 = \pi_1(S_1) \mid S_1)} (Q_2(S_2, \pi_2(S_2)) - Q_1(S_1, D_2)) \\ &+ \frac{\mathbb{I}(D_1 = \pi_1(S_1))}{\Pr(D_1 = \pi_1(S_1) \mid S_1)} \frac{\mathbb{I}(D_2 = \pi_2(S_2))}{\Pr(D_2 = \pi_2(S_2) \mid S_2)} (Y - Q_2(S_2, D_2)) \end{aligned}$$

## 6.2 Generalization to $T > 2$ Periods

The backwards recursive logic of the sequential g-formula and the doubly robust estimator generalizes directly to any finite number of time periods,  $T$ . For any sequence of adaptive policy functions  $\pi_t$ , the recursive definition of the  $Q$ -functions simply extends backwards:

$$\begin{aligned} Q_T(S_T, D_T) &= \mathbb{E}[Y \mid S_T, D_T] \\ Q_t(S_t, D_t) &= \mathbb{E}[Q_{t+1}(S_{t+1}, \pi_{t+1}(S_{t+1})) \mid S_t, D_t] \quad \text{for } t = T-1, \dots, 1 \end{aligned}$$

The final estimand is still

$$\theta = \mathbb{E}[Q_1(S_1, \pi_1(S_1))].$$

The dynamic doubly robust formula will have  $T + 1$  terms, with one term for the initial Q-function  $Q_1$  and one correction term for each of the  $T$  time periods.

However, a significant practical challenge emerges as  $T$  grows. The variance of the doubly robust estimator can increase dramatically. The correction terms involve a product of  $T$  inverse propensity scores:

$$\frac{\mathbb{I}(D_1 = \pi(S_1))}{\Pr(D_1 = \pi_1(S_1) \mid S_1)} \cdots \frac{\mathbb{I}(D_T = \pi(S_T))}{\Pr(D_T = \pi_T(S_T) \mid S_T)}$$

If the propensities are small, this product can become extremely large, leading to an estimator with very high variance. This is often called the **curse of horizon**. Even if treatments are assigned with 50% probability at each stage, the variance of the estimator will grow exponentially, proportional to  $2^T$  (since this product of propensities will grow as  $2^T$ ). This makes the estimation of long-term effects over many periods of intervention a difficult statistical problem, often requiring very large datasets or additional modeling assumptions to be feasible.

## 7 Continuous Treatments and The Dynamic DML Algorithm

The doubly robust framework is most naturally suited to discrete or low-dimensional categorical treatments, where propensity scores can be robustly estimated. When treatments are continuous (e.g., a dosage), the probability of observing any specific value  $d_t$  is zero, and the inverse propensity weighting approach breaks down.

To handle continuous treatments in dynamic settings, we can extend the Double Machine Learning (DML) framework. The key idea is to shift from modeling the full conditional outcome distribution (Q-functions) to directly modeling the treatment effect itself, typically under a partial linearity assumption.

### 7.1 Identification via Blip Functions

Instead of modeling the Q-functions, e.g.,  $\mathbb{E}[Y(d_1, d_2) \mid S_1]$  and  $\mathbb{E}[Y(d_2) \mid S_2]$ , we model the **blip functions** (or effect functions), which represents the causal effect of the treatment at time  $t$  relative to a baseline (e.g., zero treatment). For the final period, we can define the blip function as:

$$\alpha_2(S_2, d_2) := \mathbb{E}[Y(D_1, d_2) - Y(D_1, 0) \mid S_2]$$

This is the effect we would observe if we let each unit be treated based on their natural treatment in the first period, but then we intervene in the second period and we set its value to  $d_2$  or 0 and measure the difference. Here, we make a **partial linearity** assumption that the effect of the treatment  $D_2$  is linear, with a coefficient  $\delta_2$  that may depend on the state  $S_2$ . For simplicity, we

can further assume the effect is constant,

$$\alpha_2(S_2, d_2) = \delta_2 d_2.$$

This implies the conditional expectation of the outcome can be written as:

$$\mathbb{E}[Y|S_2, D_2] = \delta_2 D_2 + g_2(S_2)$$

where  $g_2(S_2)$  is some arbitrary baseline function. Hence, this is the exact analogue of the partial linearity assumption we made in the static case. Under this structure, the parameter  $\delta_2$  can be identified using the DML algorithm. We can calculate  $\delta_2$  by running a population OLS regression on the residuals:

$$Y - \mathbb{E}[Y|S_2] \quad \text{on} \quad D_2 - \mathbb{E}[D_2|S_2]$$

Once we have calculated  $\delta_2$ , we can construct an adjusted outcome that removes the effect of the observed treatment  $D_2$  and adds back the effect of our target treatment  $d_2$ :

$$Y_{adj,2} = Y - \delta_2 D_2 + \delta_2 d_2$$

This adjusted outcome represents what the outcome would have been if the second-period treatment had been set to  $d_2$ . We can then repeat this process for the first period. We assume a linear effect for the first-period treatment (when the second period is fixed to  $d_2$ ):

$$\alpha_1(S_1, d_1) := \mathbb{E}[Y(d_1, d_2) - Y(0, d_2)|S_1]$$

This is called the **blip effect** of the first period treatment. We can then make a similar linearity assumption that:

$$\alpha_1(S_1, d_1) = \delta_1 D_1$$

It can be shown that making this linearity assumption is equivalent to assuming partial linearity on the adjusted outcome, i.e.:

$$\mathbb{E}[Y_{adj,2}|S_1, D_1] = g_1(S_1) + \delta_1 D_1$$

This is the generalization of the PLM model in the dynamic regime.<sup>1</sup> We can then calculate  $\delta_1$  using the DoubleML procedure again, by running a population OLS regression on the residuals:

$$Y_{adj,2} - \mathbb{E}[Y_{adj,2} | S_1] \quad \text{on} \quad D_1 - \mathbb{E}[D_1|S_1]$$

We can then create a final adjusted outcome:

$$Y_{adj,1} = Y_{adj,2} - \delta_1 D_1 + \delta_1 d_1$$

---

<sup>1</sup>In the literature, this is referred to as a linear structural nested mean model (SNMM).

The target mean counterfactual outcome  $\mathbb{E}[Y(d_1, d_2)]$  is simply the expected value of  $Y_{adj,1}$ .

$$\mathbb{E}[Y(d_1, d_2)] = \mathbb{E}[Y_{adj,1}]$$

## 7.2 The Dynamic DML Algorithm

This gives rise to the **Dynamic DML estimation algorithm** [5].

### Algorithm: Dynamic DML for a Two-Period Model

1. **Estimate the final-stage effect,  $\delta_2$ :** Use the DML algorithm to estimate the coefficient  $\delta_2$  in the partially linear model  $\mathbb{E}[Y|S_2, D_2] = g_2(S_2) + \delta_2 D_2$ . This involves cross-fitted nuisance regressions for  $\mathbb{E}[Y|S_2]$  and  $\mathbb{E}[D_2|S_2]$ . Let the estimate be  $\hat{\delta}_2$ .

2. **Generate first-stage pseudo-outcomes:** For each observation  $i$ , compute the adjusted outcome:

$$\hat{Y}_{i,adj,2} = Y_i - \hat{\delta}_2 D_{2,i} + \hat{\delta}_2 d_2$$

3. **Estimate the first-stage effect,  $\delta_1$ :** Use the DML algorithm to estimate the coefficient  $\delta_1$  in the model  $\mathbb{E}[\hat{Y}_{adj,2}|S_1, D_1] = g_1(S_1) + \delta_1 D_1$ . This uses the pseudo-outcomes from the previous step as the dependent variable.

4. **Generate final adjusted outcomes:** For each observation  $i$ , compute the final prediction:

$$\hat{Y}_{i,adj,1} = \hat{Y}_{i,adj,2} - \hat{\delta}_1 D_{1,i} + \hat{\delta}_1 d_1$$

5. **Estimate the mean potential outcome:** The final estimate is the sample average of these predictions:

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n \hat{Y}_{i,adj,1}$$

This algorithm provides a powerful way to handle continuous treatments in dynamic settings, avoiding the variance issues of inverse propensity weighting while still allowing for flexible machine learning models to control for the confounding effects of the state variables.

## 7.3 Dynamic DML as a Moment-Based Estimator

A more formal way to view the Dynamic DML algorithm is as a method for solving a system of moment equations. This perspective, grounded in the work of Chernozhukov et al. (2018) [6], is what allows us to derive standard errors and confidence intervals for our estimates.

We can view this algorithm as solving the empirical version of the normal equations of the

corresponding OLS problems at each of the two steps:

$$\begin{aligned}\mathbb{E}[(\tilde{Y}_2 - \delta_2 \tilde{D}_2) \tilde{D}_2] &= 0 \\ \mathbb{E}[(\tilde{Y}_1 - \delta_1 \tilde{D}_1) \tilde{D}_1] &= 0\end{aligned}$$

The target quantity  $\tau = \mathbb{E}[Y(d_1, d_2)]$  is given as a function of  $\delta_1, \delta_2$ :

$$\tau = \mathbb{E}[Y - \delta_2 D_2 + \delta_2 d_2 - \delta_1 D_1 + \delta_1 d_1]$$

Note that  $\tilde{Y}_1$  implicitly depends on  $\delta_2$ , since it is the residual of the adjusted outcome. We make it explicit. By the linearity of the residualization operator, we can write:

$$\tilde{Y}_1 = \tilde{Y}^{(1)} - \delta_2 \tilde{D}_2^{(1)}, \quad \tilde{V}^{(1)} = V - \mathbb{E}[V \mid S_1]$$

Thus we can re-write the system of normal equations as:

$$\begin{aligned}\mathbb{E}[(\tilde{Y}_2 - \delta_2 \tilde{D}_2) \tilde{D}_2] &= 0 \\ \mathbb{E}[(\tilde{Y}^{(1)} - \delta_2 \tilde{D}_2^{(1)} - \delta_1 \tilde{D}_1) \tilde{D}_1] &= 0\end{aligned}$$

Both of these equations are insensitive to the nuisance functions, which correspond to the regression problems:  $\mathbb{E}[Y \mid S_2]$ ,  $\mathbb{E}[Y \mid S_1]$ ,  $\mathbb{E}[D_2 \mid S_2]$ ,  $\mathbb{E}[D_2 \mid S_1]$ ,  $\mathbb{E}[D_1 \mid S_1]$ .

Thus we have concluded that the parameters  $\tau, \delta_1, \delta_2$  are solutions to the following system of equations, which are insensitive to nuisance functions

$$\begin{aligned}\mathbb{E}[(\tilde{Y}_2 - \delta_2 \tilde{D}_2) \tilde{D}_2] &= 0 \\ \mathbb{E}[(\tilde{Y}^{(1)} - \delta_2 \tilde{D}_2^{(1)} - \delta_1 \tilde{D}_1) \tilde{D}_1] &= 0 \\ \mathbb{E}[Y - \delta_2 D_2 + \delta_2 d_2 - \delta_1 D_1 + \delta_1 d_1 - \tau] &= 0\end{aligned}$$

The estimated parameters  $\hat{\tau}, \hat{\delta}_1, \hat{\delta}_2$  based on the dynamic DML algorithm are the solution to the empirical analogue of these equations, where we use cross-fitted estimates of the nuisance functions. We now give a small detour to a general statement about such parameters defined as solutions to insensitive moment equations.

### 7.3.1 Generic Moment Equation Based Estimator

Suppose that a vector of parameters of interest  $\theta_0$  is the solution to a vector of equations:

$$\theta_0 \text{ is solution to } \mathbb{E}[m(Z; \theta, g_0)] = 0$$

for some known *moment functions*  $m$  that depend on nuisance functions  $g_0$  but are insensitive to the nuisance functions. Consider the estimated parameters  $\hat{\theta}$  that are the solution to the empirical

plug-in analogue of these equations:

$$\hat{\theta} \text{ is solution to } \mathbb{E}_n[m(Z; \theta, \hat{g})] = 0$$

where nuisances  $\hat{g}$  are estimated in a cross-fitting manner.

#### Generic Asymptotic Normality (Chernozhukov et al 2018) [6]

Under regularity conditions and sufficiently fast RMSE rates ( $\ll n^{-1/4}$ ) on the nuisances, the parameters  $\hat{\theta}$  are jointly asymptotically normal:

$$\sqrt{n}(\hat{\theta} - \theta) \rightarrow N(0, J^{-1} \Sigma J^{-\top})$$

where  $J$  is the Jacobian of the equations with respect to the parameters:

$$J = \nabla_{\theta} \mathbb{E}[m(Z; \theta_0, g_0)]$$

and  $\Sigma$  is the covariance matrix of the equations:

$$\Sigma = \mathbb{E}[m(Z; \theta_0, g_0)m(Z; \theta_0, g_0)']$$

Standard errors can be calculated as  $\sqrt{\hat{V}/n}$  using estimated variance  $\hat{V} = \hat{J}^{-1} \hat{\Sigma} \hat{J}^{-\top}$ :

$$\hat{J} = \nabla_{\theta} \mathbb{E}_n[m(Z; \hat{\theta}, \hat{g})] \quad \hat{\Sigma} = \mathbb{E}_n[m(Z; \hat{\theta}, \hat{g})m(Z; \hat{\theta}, \hat{g})']$$

### 7.3.2 Confidence Intervals for Dynamic DML Algorithm

The parameters of interest are  $\theta = (\tau, \delta_1, \delta_2)$ , where  $\tau = \mathbb{E}[Y(d_1, d_2)]$  is our target, and  $\delta_1, \delta_2$  are the linear blip effects. As we have argued, they can be viewed as the solution to the following equations

$$\mathbb{E}[m(Z; \theta, g_0)] = 0 \quad m(Z; \theta, g_0) = \begin{pmatrix} (\tilde{Y}_2 - \delta_2 \tilde{D}_2) \tilde{D}_2 \\ (\tilde{Y}^{(1)} - \delta_2 \tilde{D}_2^{(1)} - \delta_1 \tilde{D}_1) \tilde{D}_1 \\ Y - \delta_2 D_2 + \delta_2 d_2 - \delta_1 D_1 + \delta_1 d_1 - \tau \end{pmatrix}$$

which are insensitive to nuisance functions  $g_0$ , where  $g_0$  are the regression functions:  $\mathbb{E}[Y | S_2]$ ,  $\mathbb{E}[Y | S_1]$ ,  $\mathbb{E}[D_2 | S_2]$ ,  $\mathbb{E}[D_2 | S_1]$ ,  $\mathbb{E}[D_1 | S_1]$ . The estimated parameters  $\hat{\tau}, \hat{\delta}_1, \hat{\delta}_2$  based on the dynamic DML algorithm are the solution to the empirical analogue of these equations for all parameters (with cross-fitting for the nuisances). Thus, **we can apply the general framework to these equations to derive confidence intervals and standard errors.**

## Dynamic DML: Pseudocode with Standard Errors

```
def dynamic_dml(Y, D1, D2, S1, S2, d1, d2, K=5):
    n = len(Y)
    # Initialize cross-fitted nuisance predictions
    E_Y_S2, E_D2_S2 = np.zeros(n), np.zeros(n)
    E_Y_S1, E_D2_S1, E_D1_S1 = np.zeros(n), np.zeros(n), np.zeros(n)
    kf = KFold(n_splits=K, shuffle=True)
    for train, test in kf.split(S1):
        # Nuisance models (e.g., GradientBoostingRegressor)
        E_Y_S2[test] = ML().fit(S2[train], Y[train]).predict(S2[test]) # E[Y|S2]
        E_D2_S2[test] = ML().fit(S2[train], D2[train]).predict(S2[test]) # E[D2|S2]
        E_Y_S1[test] = ML().fit(S1[train], Y[train]).predict(S1[test]) # E[Y|S1]
        E_D2_S1[test] = ML().fit(S1[train], D2[train]).predict(S1[test]) # E[D2|S1]
        E_D1_S1[test] = ML().fit(S1[train], D1[train]).predict(S1[test]) # E[D1|S1]

    # Construct residuals:  $\tilde{V} = V - E[V|S_t]$ 
    Y_t2 = Y - E_Y_S2; D2_t2 = D2 - E_D2_S2 # Stage 2 residuals
    Y_t1 = Y - E_Y_S1; D2_t1 = D2 - E_D2_S1 # Stage 1 residuals of Y, D2
    D1_t1 = D1 - E_D1_S1 # Stage 1 residual of D1

    # Closed-form solutions to  $E[m(Z; \theta, g_0)] = 0$ 
    delta2 = np.mean(Y_t2 * D2_t2) / np.mean(D2_t2**2)
    delta1 = np.mean((Y_t1 - delta2*D2_t1) * D1_t1) / np.mean(D1_t1**2)
    tau = np.mean(Y - delta2*D2 + delta2*d2 - delta1*D1 + delta1*d1)

    # Per-observation moment values  $m_i(\theta)$ 
    m1_i = (Y_t2 - delta2 * D2_t2) * D2_t2
    m2_i = (Y_t1 - delta2*D2_t1 - delta1*D1_t1) * D1_t1
    m3_i = Y - delta2*D2 + delta2*d2 - delta1*D1 + delta1*d1 - tau

    # Jacobian  $J = E_n[\text{nabla}_{\theta} m(Z; \theta, g_0)]$ 
    J = np.array([[ -np.mean(D2_t2**2), 0, 0],
                  [ -np.mean(D2_t1 * D1_t1), -np.mean(D1_t1**2), 0],
                  [ np.mean(d2 - D2), np.mean(d1 - D1), -1]])

    # Sandwich variance:  $V = J^{-1} \Sigma J^{-T} / n$ 
    M = np.column_stack([m1_i, m2_i, m3_i])
    Sigma = M.T @ M / n
    J_inv = np.linalg.inv(J)
    V = J_inv @ Sigma @ J_inv.T / n
    se_delta2, se_delta1, sp_tau = np.sqrt(np.diag(V))

    return tau, delta1, delta2, sp_tau, se_delta1, se_delta2
```

## 8 Summary and Takeaways

This chapter introduced the fundamental concepts for analyzing causal effects in settings with sequential, time-varying treatments. The presence of time-varying confounders that are also affected by past treatments breaks standard identification strategies. The key ideas for overcoming this challenge are:



1. **Time-varying confounding is the central challenge.** Intermediate variables that are both mediators of past treatments and confounders of future treatments (e.g.,  $S_2$ ) prevent identification via simple conditioning. We must use a dynamic approach.
2. **Backwards recursion provides the solution.** The **sequential g-formula** identifies the effect of a joint intervention by recursively solving the problem backwards in time. It first identifies the effect of the last treatment conditional on the preceding state, then uses this to form an adjusted outcome for identifying the effect of the next-to-last treatment, and so on.
3. **Doubly robust methods are essential for inference.** To use flexible machine learning models for estimation while retaining valid confidence intervals, we must use a **dynamic doubly robust** (or multiply robust) estimator. This involves adding a correction term for each time period, which uses both Q-function and propensity score models.
4. **The framework supports adaptive policies and continuous treatments.** The g-formula and its doubly robust counterpart can be easily adapted to evaluate complex, state-dependent **adaptive policies**. For continuous treatments, the **Dynamic DML algorithm** extends the logic of partial linearity and residualization to the sequential setting.
5. **Long horizons pose a statistical challenge.** While the framework generalizes to many time periods, the variance of estimators can grow exponentially with the time horizon (the “curse of horizon”), making long-term effect estimation statistically demanding.

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